



RAJARATA UNIVERSITY OF SRI LANKA

FACULTY OF TECHNOLOGY

Bachelor of Information and Communication Technology
First Year - Semester I Examination – September/October 2022

ICT 1202 – ELECTRONIC CIRCUITS

Time: Two (02) hours

Answer all Questions

1.

a) Describe the concept of Analog electronics. [10 marks]

b) Prove the following Boolean equities using Boolean theorems and identities.

i. $[AB'(C+BD) + A'B']C$

ii. $\overline{AB + AC} + \overline{ABC}$

[24 marks]

c) Students of Rajarata University have implemented an automatic ground watering system to eliminate the manual watering procedure which was not efficient. The newly implemented system takes input readings from four sensor units, A, B, C, and D (Figure 1.1), and based on their values the water pump is activated through a control unit. All four sensor units are capable of producing only two states that are active and inactive. It is said that the water pump's activation X, is given in the following Boolean algebraic statement:

$$X = \overline{A}\overline{B}\overline{C}D + \overline{A}\overline{B}CD + \overline{A}B\overline{C}\overline{D} + \overline{A}B\overline{C}D + \overline{A}BC\overline{D} + \overline{A}BCD + A\overline{B}\overline{C}\overline{D} + ABC\overline{D}$$

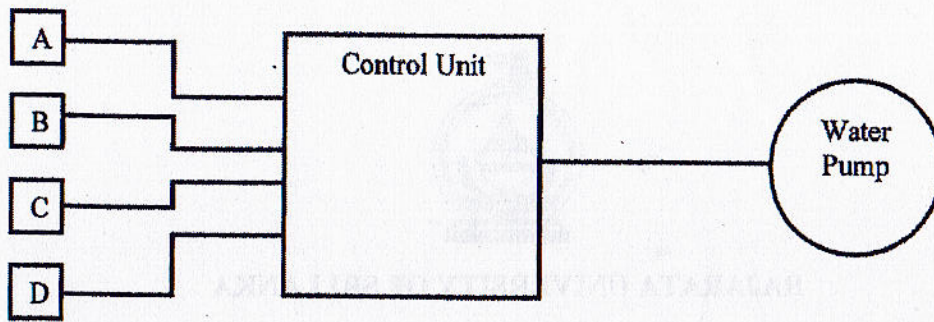


Figure 1.1

- i. Complete the truth table given below for the above Boolean logic expression:

A	B	C	D	X
0	0	0	0	
0	0	0	1	
0	0	1	0	
0	0	1	1	
0	1	0	0	
0	1	0	1	
0	1	1	0	
0	1	1	1	
1	0	0	0	
1	0	0	1	
1	0	1	0	
1	0	1	1	
1	1	0	0	
1	1	0	1	
1	1	1	0	
1	1	1	1	

- ii. In order to design the most simplified circuit with a minimum number of logic gates, draw the Karnaugh-Map for the above (a) truth table and by clearly showing your grouping, derive the simplified Boolean expression in both Sum of Product and Product of Sum forms.
- iii. Design the logic circuits for the Sum of Product and Product of Sum expressions in above (ii) using only NAND and NOR gates. Clearly indicate the method and steps.
- [51 marks]**

- d) (Figure 1.2) A lamp outside a front door comes ON automatically when it is dark, and someone stands on the doormat outside the front door.
 A pressure sensor under the doormat changes from OFF (0) to ON (1) when someone stands on doormat.
 The light sensor is ON (1) when it is light and OFF when it is dark.
 The block diagram of the circuits is shown below.

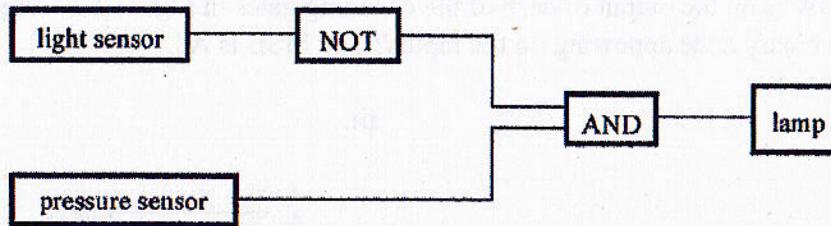


Figure 1.2

When the system was first set up, light from the lamp could shine onto the light sensor. What would happen when someone stood on the doormat when it was dark? Explain your answer.

[15 marks]

[Total 100 marks]

2.

a) List down three functions of combinational logic.

[9 marks]

b) Implement the expression using a multiplexer.

$$F(A, B, C) = \sum m(2, 3, 6, 7)$$

[37 marks]

c) Fill in the following truth table using the given circuit (Figure 2.1). Ignore rows not included in the table.

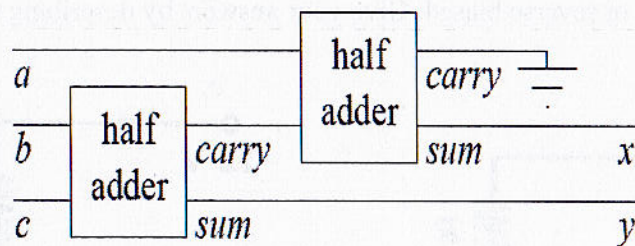


Figure 2.1

<i>a</i>	<i>b</i>	<i>c</i>	<i>x</i>	<i>y</i>
0	1	1		
1	0	1		
1	1	0		
1	1	1		

[24 marks]

- d) What is the meaning of “Glitch”? Explain it. [10 marks]
 e) When a LOW is on the output of each of the decoding gates in Figure 2.2 – Figure 2.5, what is the binary code appearing on the inputs? The MSB is A_3 .

i.

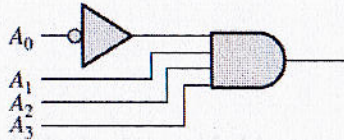


Figure 2.2

ii.

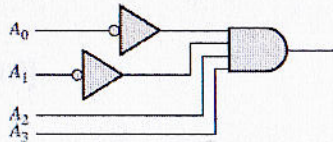


Figure 2.3

iii.

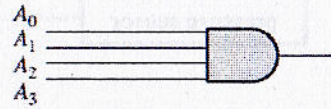


Figure 2.4

iv.

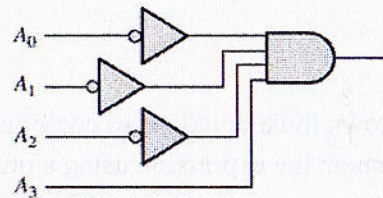


Figure 2.5

[20 marks]

[Total 100 marks]

3.

- a) In the following figures (Figure 3.1, Figure 3.2), identify whether these diodes are forward biased or reverse biased. Give your answers by describing your reasons.

i.

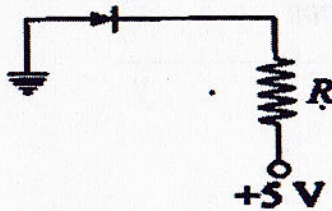


Figure 3.1

ii.

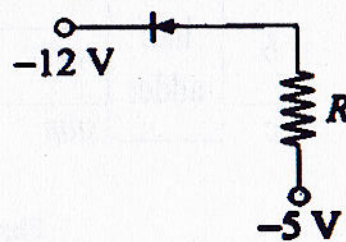


Figure 3.2

[30 marks]

- b) Find the α rating of the transistor shown in following Figure 3.3. Hence determine the value of I_C using both α and β rating of the transistor.

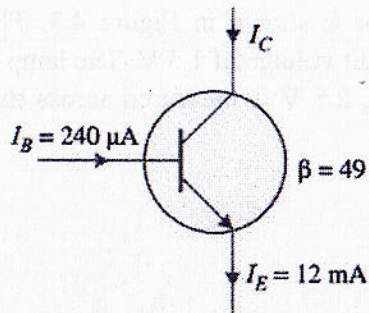


Figure 3.3

[30 marks]

- c) In a common base connection, $\alpha = 0.95$. The voltage drops across $2\text{ k}\Omega$ resistor which is connected in the collector is 2 V . Find the base current. (Figure 3.4)

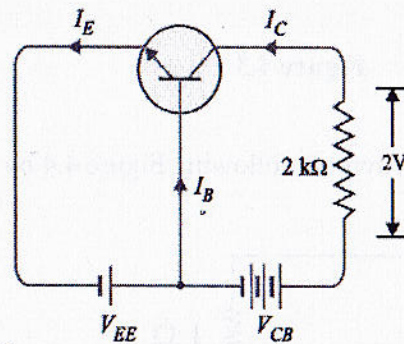


Figure 3.4

[40 marks]

[Total 100 marks]

4.

- a) Find the equivalent resistance at the indicated terminal pair for each of the networks shown in Figure 4.1 and Figure 4.2.

i)

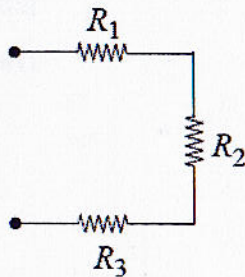


Figure 4.1

ii)

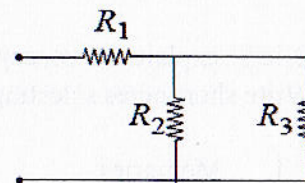


Figure 4.2

[20 marks]

- b) A pictorial diagram for a flashlight is shown in Figure 4.3. The two batteries are identical, and each has an open-circuit voltage of 1.5 V. The lamp has a resistance of $5\ \Omega$ when lit. With the switch closed, 2.5 V is measured across the lamp. What is the internal resistance of each battery?

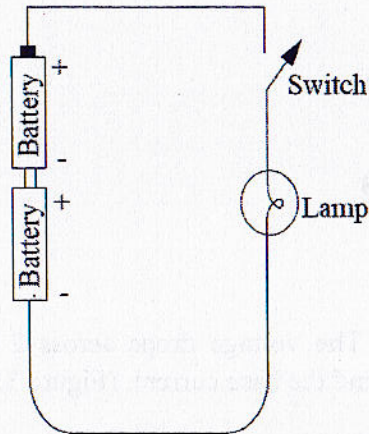


Figure 4.3

[20 marks]

- c) Determine the current i_0 in the circuit in following Figure 4.4 by working with resistors in series and parallel.

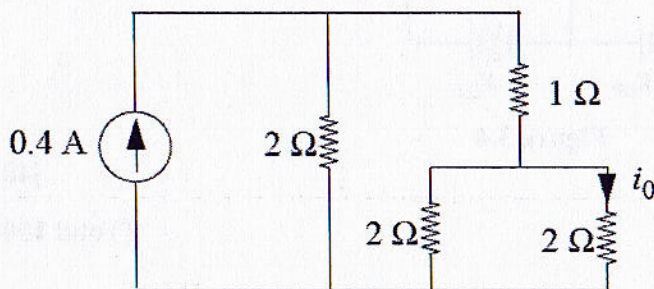


Figure 4.4

[20 marks]

- d) Briefly explain the concept of sequential logic.
e) Write short notes selecting two topics from the following.

[10 marks]

- i. Memories
- ii. Timers
- iii. Counters
- iv. Shift Registers

[30 marks]

[Total 100 marks]

--End--